In [1]:
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# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_diabetes
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load the diabetes dataset
diabetes = load_diabetes()

# Create DataFrame
df = pd.DataFrame(diabetes.data, columns=diabetes.feature_names)

# Independent variable (X)
X = df[['bmi']]

# Dependent variable (y)
y = diabetes.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Print results
print("Coefficient (Slope):", model.coef_[0])
print("Intercept:", model.intercept_)

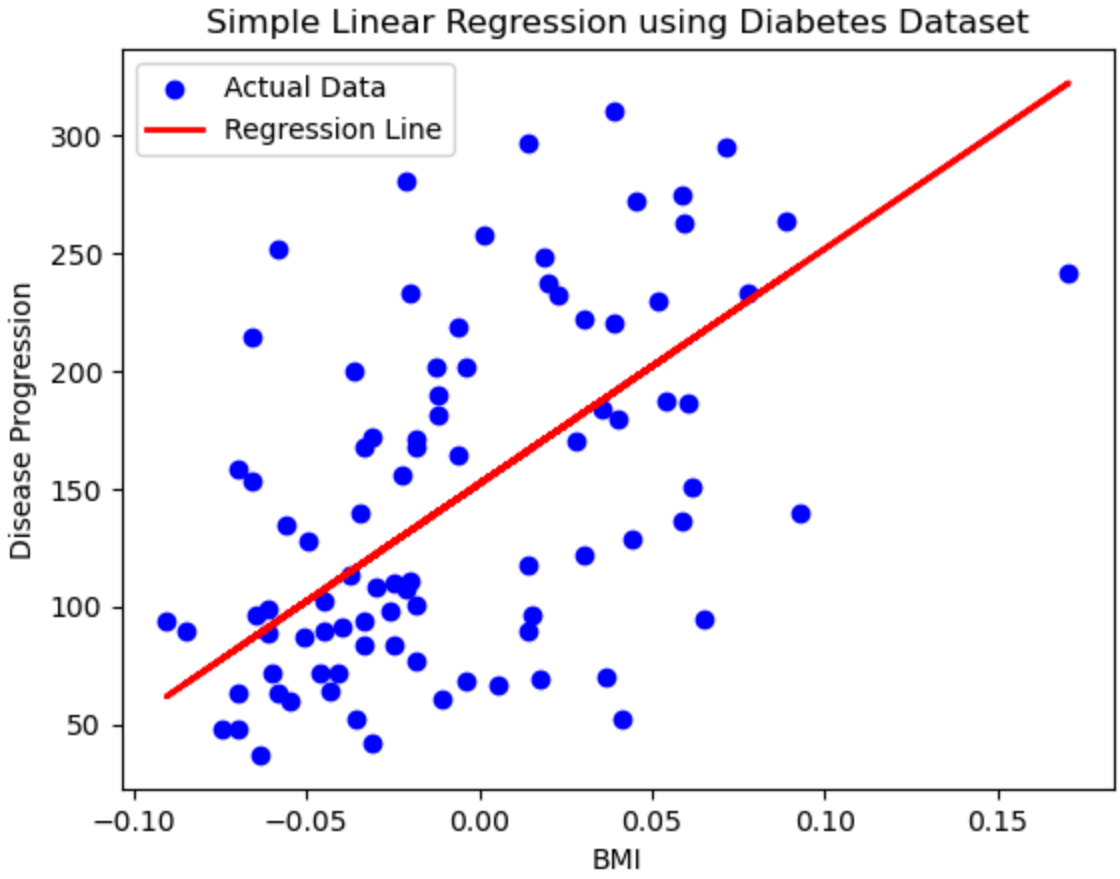
# Evaluation metrics
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R² Score:", r2_score(y_test, y_pred))

# Plot the regression line
plt.scatter(X_test, y_test, color='blue', label='Actual Data')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line')

# Labels and title
plt.xlabel('BMI')
plt.ylabel('Disease Progression')
plt.title('Simple Linear Regression using Diabetes Dataset')
plt.legend()
```

```
# Show plot
plt.show()
```

```
Coefficient (Slope): 998.5776891375593
Intercept: 152.00335421448167
Mean Squared Error: 4061.8259284949268
R² Score: 0.23335039815872138
```



In [ ]: